

Data Encryption Standard (DES) & Public Key Cryptography

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What is Cryptography?

- Cryptography is the art of securing information by transforming it into an unreadable format.
- It ensures confidentiality, integrity, and authenticity of data.
- Key terms: plaintext, ciphertext, encryption, decryption, cipher.

Overview of DES

- DES (Data Encryption Standard) is a symmetric-key algorithm for the encryption of digital data.
- Standardized in the 1970s by NIST.
- Uses a 56-bit key and processes 64-bit blocks.
- Commonly used for securing sensitive data.

Feistel Cipher Structure

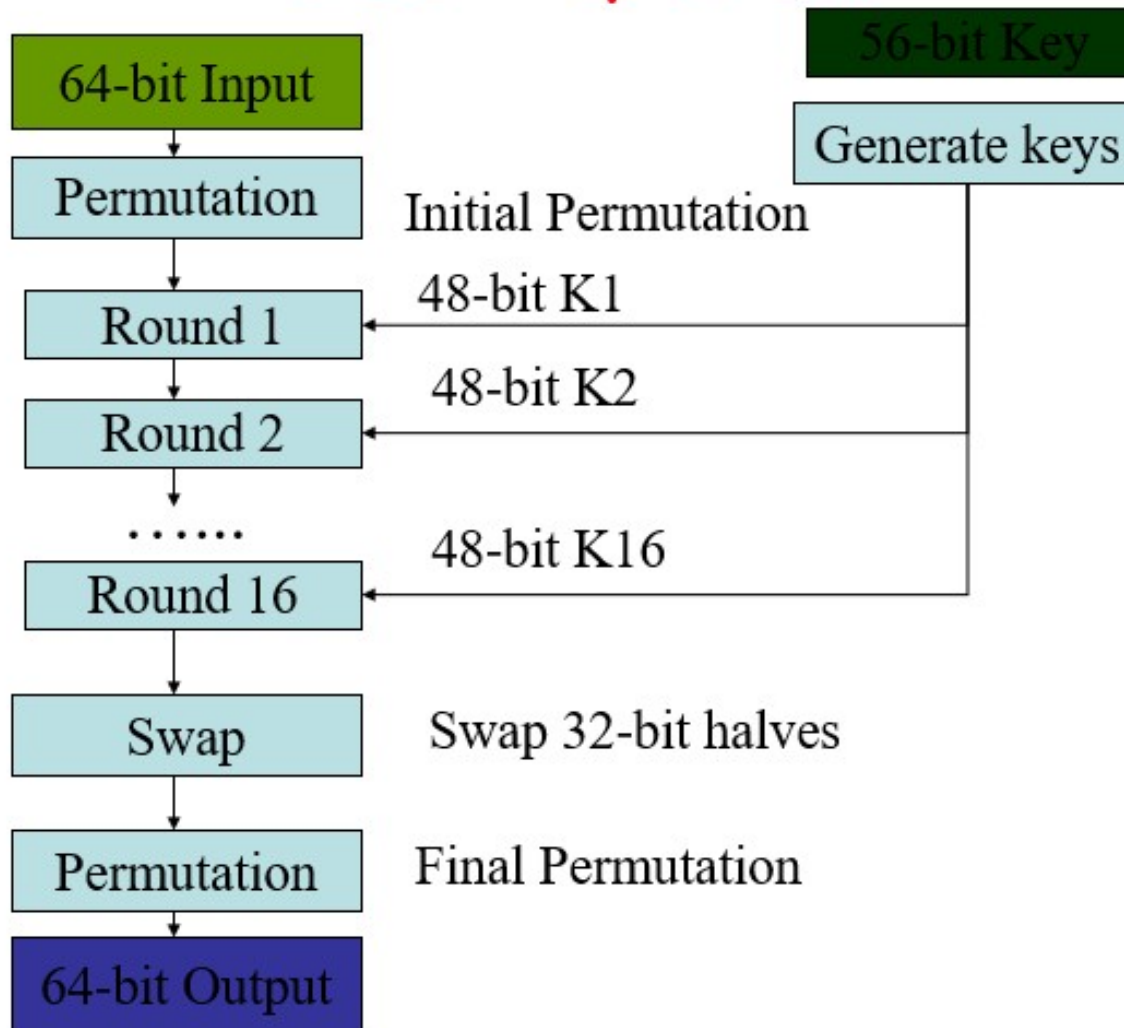


- DES is based on the Feistel structure.
- Each round splits the data into two halves and processes them through a function.
- The structure allows encryption and decryption with the same algorithm.

DES Working – Top View

- Begins with an Initial Permutation (IP).
- 16 rounds of processing with subkeys.
- Ends with an Inverse Initial Permutation.
- Subkeys are generated using a key schedule algorithm.

DES Top View



Plaintext (64 bits):

011011010010110001010111011110011001101010111100
1101111011110001

Key (56 bits):

101010101011101100001001000110010001010101011110

Step 1: Initial Permutation (IP)

Data bits are shuffled according to a fixed table.

After IP:

L0 = 01101101001011000101011101111001

R0 = 10011010101111001101111011110001

Step 2: Round Function (f)

a) E-Box Expansion (32-bit to 48-bit)

$R_0 = 10011010101111001101111011110001$

$E(R_0) = 110010\ 011010\ 101011\ 110011\ 011110$
 $111100\ 011000$

b) XOR with Round Key (48-bit)

$E(R_0) \text{ XOR } K_1$

c) S-Box Substitution Divide 48 bits into eight 6-bit chunks and substitute using S-boxes.

Example: $110010 \rightarrow 1010$

d) P-Box Permutation Rearrange the 32 bits using a P-box table.

$f(R0, K1) = 01101111010111000101101011010101$

Step 3: XOR with L0

$L0 \text{ XOR } f(R0, K1) =$

00000010011100000000110110101100

Step 4: Swap Halves

$L1 = R0$

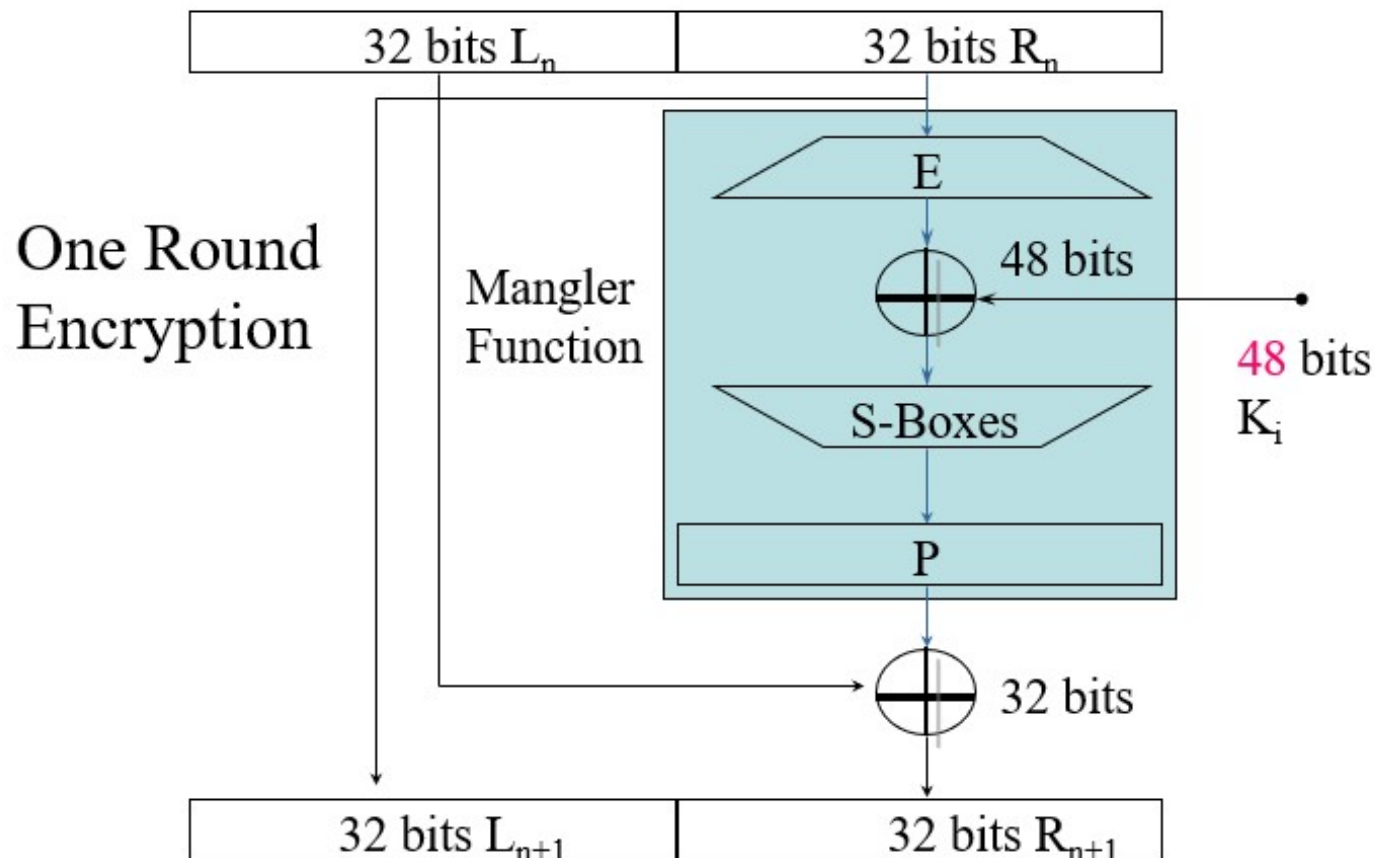
$R1 = \text{New R}$

DES Round Details



- Each round involves expansion, key mixing, substitution, and permutation.
- Expansion (E-box) expands 32 bits to 48 bits.
- Substitution uses 8 S-boxes to reduce back to 32 bits.
- Final permutation (P-box) rearranges bits.

A DES Round



- A small change in input causes a large change in output.
- DES achieves a good avalanche effect, increasing security.
- Ensures that patterns in plaintext do not show in ciphertext.

Weaknesses of DES

- 56-bit key is too short by modern standards.
- Vulnerable to brute-force attacks.
- EFF cracked DES in 1998 using custom hardware.

Triple DES & AES

- Triple DES (3DES) applies DES three times for improved security.
- AES (Advanced Encryption Standard) replaced DES in 2001.
- AES supports 128, 192, and 256-bit keys and 128-bit blocks.